

Climate
change.





GLOBAL WARMING

global warming.
oh no.

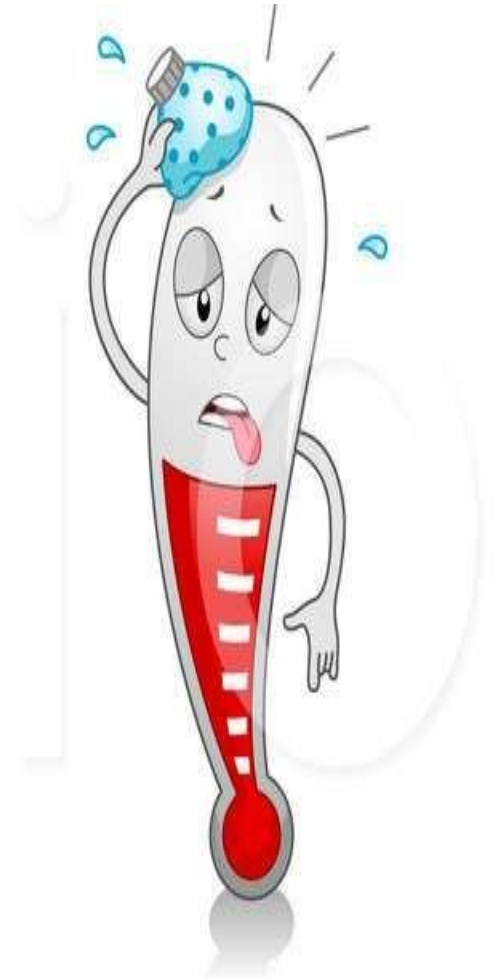


GLOBAL WARMING is an increase in the earth's average atmospheric temperature that causes corresponding changes in climate and that may result from the greenhouse effect.



MORE ABOUT GLOBAL WARMING

- ❖ A gradual increase in the overall temperature of the earth's atmosphere generally attributed to the greenhouse effect caused by increased levels carbon dioxide, CFCs, and other pollutants.
- ❖ Since the early 20th century, Earth's mean surface temperature has increased by about 0.8 °C (1.4 °F), with about two-thirds of the increase occurring since 1980.
- ❖ Warming of the climate system is unequivocal, and scientists are 95-100% certain that it is primarily caused by increasing concentrations of greenhouse gases produced by human activities such as the burning of fossil fuels and deforestation.



Causes of Global Warming

- ❖ Global warming is primarily a problem of too much carbon dioxide (CO₂) in the atmosphere—which acts as a blanket, trapping heat and warming the planet.
- ❖ As we burn fossil fuels like coal, oil and natural gas for energy or cut down and burn forests to create pastures and plantations, carbon accumulates and overloads our atmosphere.
- ❖ Certain waste management and agricultural practices aggravate the problem by releasing other potent global warming gases, such as methane and nitrous oxide.



Cont....

- ❖ The major greenhouse gases are water vapor, which causes about 36–70% of the greenhouse effect; carbon dioxide (CO_2), which causes 9–26%; methane (CH_4), which causes 4–9%; and ozone (O_3), which causes 3–7%. Clouds also affect the radiation balance.
- ❖ Human activity since the Industrial Revolution has increased the amount of greenhouse gases in the atmosphere, leading to increased radiative forcing from CO_2 , methane, tropospheric ozone, CFCs and nitrous oxide.

GLOBAL WARMING

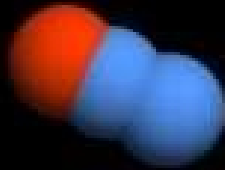


Effects of global

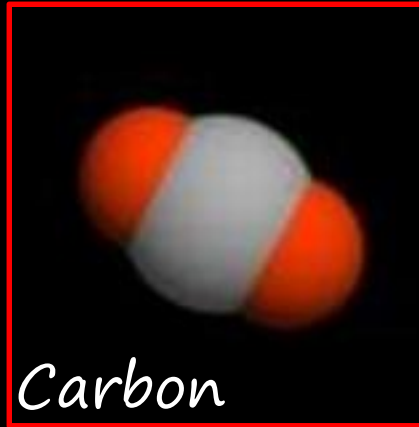
- ❖ Sea level rise is accelerating. The planet's temperature is rising.
- ❖ The number of large wildfires is growing.
- ❖ Dangerous heat waves are becoming more common.
- ❖ Extreme storm events are increasing in many areas. More severe droughts are occurring in others.
- ❖ These are having significant and harmful effects on our health, our environment, and our communities.



Greenhouse Gases



*Nitrous
oxide*



*Carbon
dioxide*



*Methan
e*

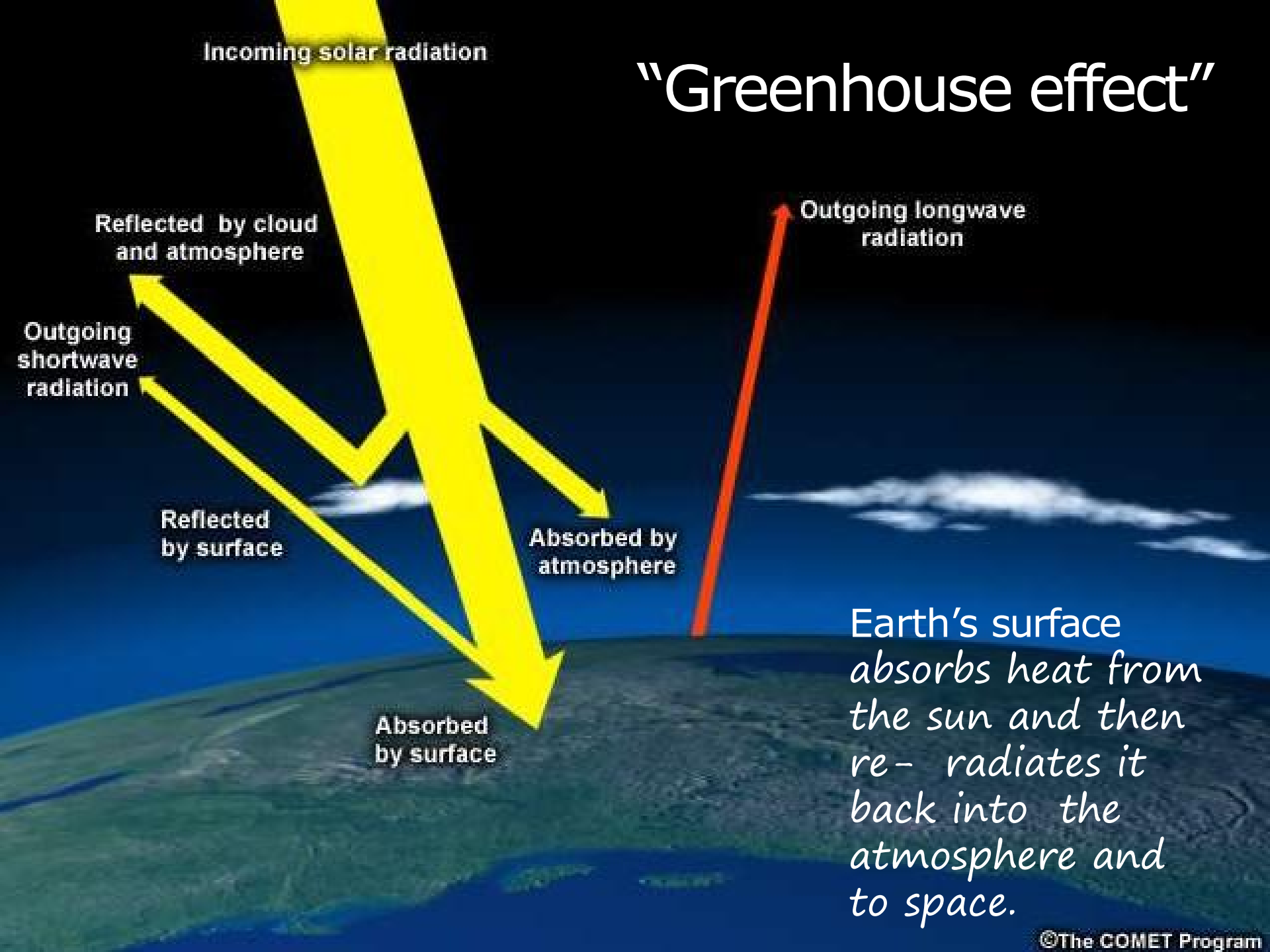


Water



*Sulfur
hexafluoride*

"Greenhouse effect"



Earth's surface absorbs heat from the sun and then re-radiates it back into the atmosphere and to space.

Increasing greenhouse gases trap more heat

Much of this heat is absorbed by greenhouse gases, which then send the heat back to the surface, to other greenhouse gas molecules, or out to space. Though only 1% of atmospheric gases are greenhouse gases, they are extremely powerful heat trappers. By burning fossil fuels faster and faster, humans are effectively piling on more blankets, heating the planet so much and so quickly that it's

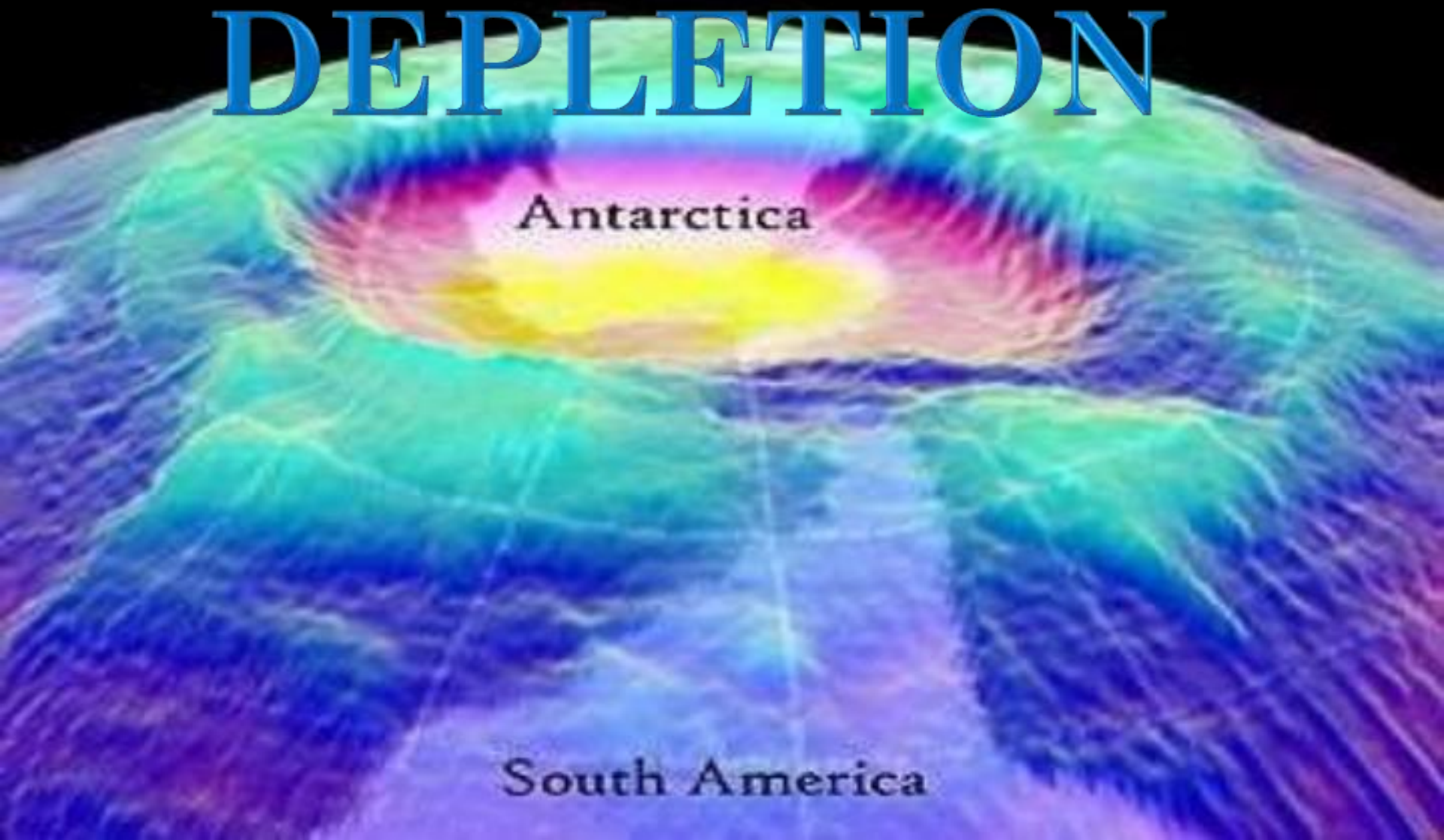


How to reduce Global Warming?

- ❖ Produce more fuel-efficient vehicles
- ❖ Reduce vehicle use
- ❖ Improve energy-efficiency in buildings
- ❖ Develop carbon capture and storage processes
- ❖ Triple nuclear power
- ❖ Increase solar power
- ❖ Decrease deforestation/plant forests



OZONE LAYER DEPLETION



- ❖ The **ozone layer** in the upper atmosphere absorbs the sun's ultraviolet radiation, preventing it from reaching the earth's surface.
- ❖ This layer in the atmosphere protects life on earth from the dangerous UV radiation from the sun.
- ❖ In the 1970s, scientists discovered that chemicals called chlorofluorocarbons or CFCs, which were used as refrigerants and aerosol spray propellants, posed a threat to the ozone layer.
- ❖ The destruction of the ozone layer is seen to cause increased cases of skin cancer and cataracts.
- ❖ It also causes damage to certain crops and to plankton, thus affecting nature's food chains and food webs. This in turn causes an increase in carbon dioxide due to the decrease in vegetation.





What is acid

ACID RAIN is a rain
any other form
precipitation
usually acidic i.e. it has
low pH. Acid rain is
caused by emissions
of sulphur dioxide and
nitrogen dioxide, which
react with water in the
atmosphere to produce
acids.



Causes of Acid

- ❖ The primary cause of acid rain is the emission of harmful gases and emissions by industrial power plants and factories, and to some extent, cars as well.
- ❖ Burning of fossil fuels lead to emission of gases that give oxides of sulfur and nitrogen.
- ❖ Burning of gasoline and diesel in cars build the amount of acidified water droplets in the atmosphere.



Effects of Acid

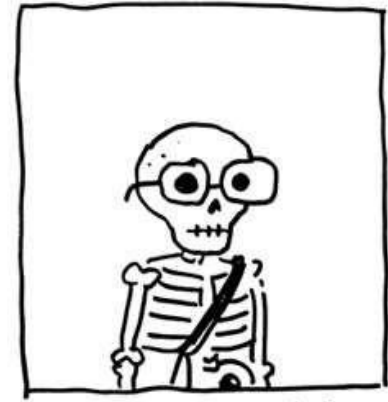
- ❖ The effects of acid rain are unhealthy and harmful. Buildings, national monuments and bridges tend to deteriorate faster.
- ❖ The biggest reason to find satisfactory solutions for acid rains is the damage to natural ecosystems. This rain will affect the plants, trees, rivers, lakes, soil and land wherever it may fall on.
- ❖ Animals and plants that live in such areas are severely affected. Their food sources get diminished, they may develop unhealthy diseases or mutations, or they may even die in extreme cases.



EFFECT ON HUMANS

The pollutants that cause acid rain—sulfur dioxide and nitrogen oxides damage human health.

These gases interact in the atmosphere to form fine sulfate and nitrate particles that can be transported long distances by winds and inhaled deep into people's lungs. When we breathe in air pollution, these very fine particulates can easily enter our body, where they can cause breathing problems (asthma), bronchitis and over time even cause skin



Solutions for acid

- **Understand acid deposition's causes and effects**

To solve the acid rain problem, people need to understand how acid rain damages the environment. They also need to understand what changes could be made to the air pollution sources that cause the problem.

- **Use Low Sulfur Coal**

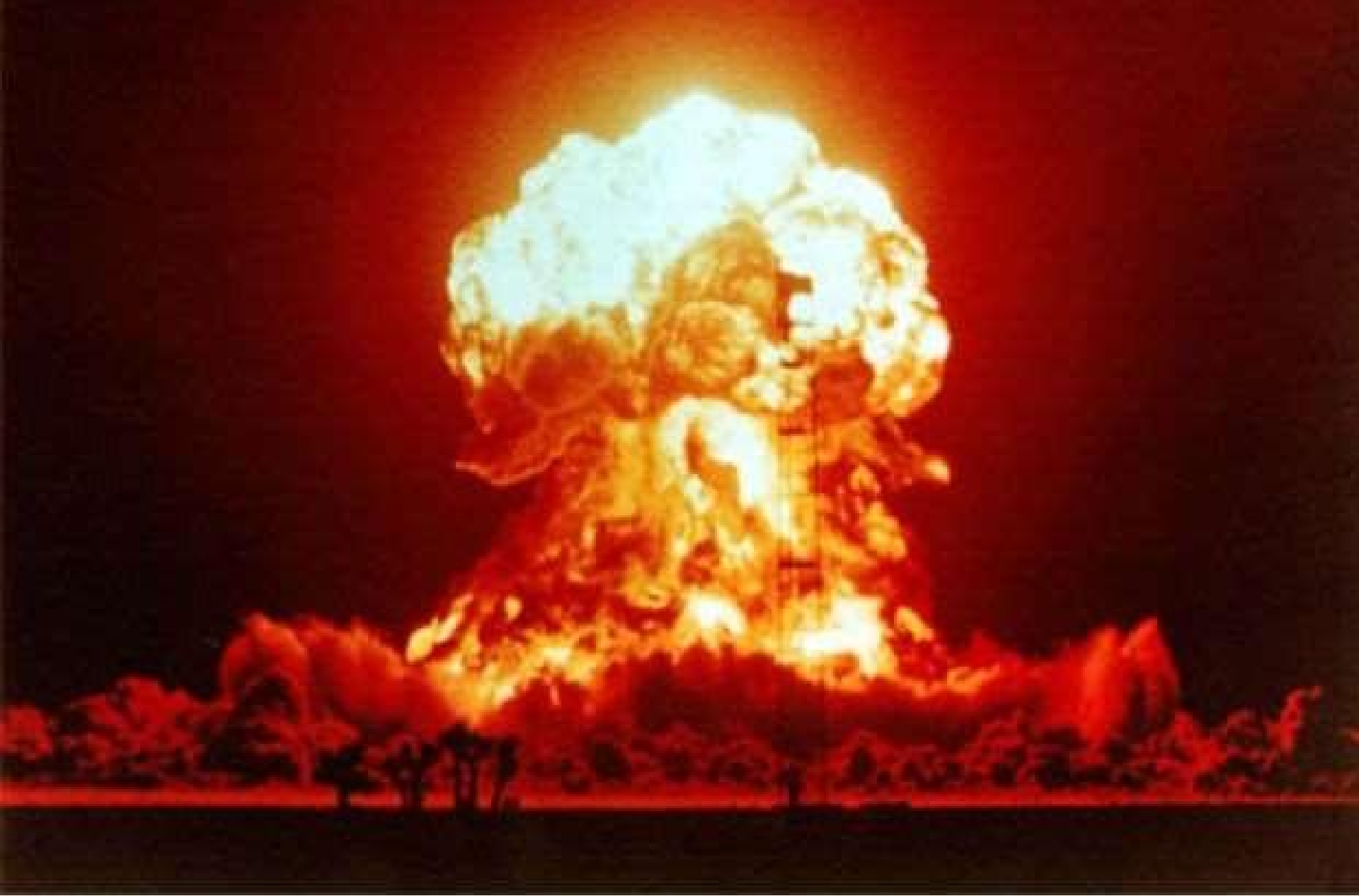
The amount of sulfur oxides emitted by a power plant can be reduced by burning coal with a very low percentage of sulfur present.

- **Use Scrubbers**

Scrubbers remove 80–95 percent of the sulfur oxides. Certain types do not remove nitrogen oxides. They are costly to retrofit to existing power plants. They increase the electric generating costs by 10–15 %.



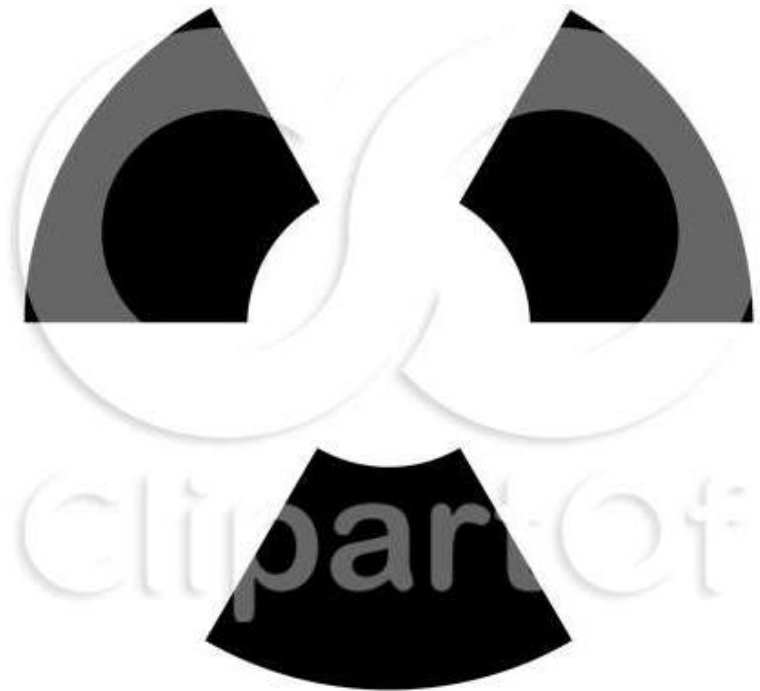
NUCLEAR DISASTERS



What are Nuclear Disasters?

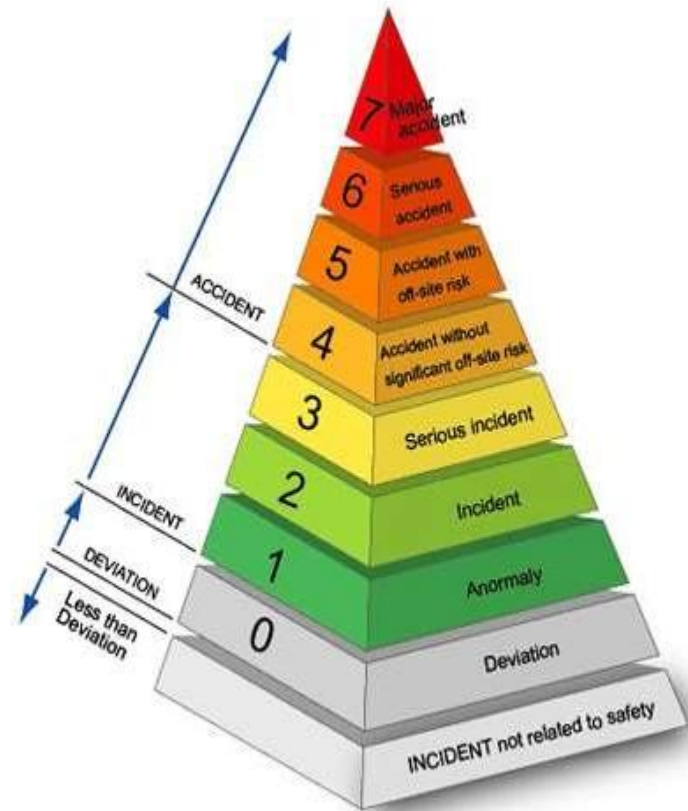
Nuclear energy is generated by the process of molecular fission, molecular fusion and molecular decay. Uncontrolled formation of such energy leads to nuclear disasters.

These disasters significantly affect the population and its effects are observed for years.



Effects on Environment

- ❖ A single nuclear accident can cause loss of life, long term illness and destruction of property on a large scale.
- ❖ Radioactivity leads to cancer, genetic disorders and death in the affected area for decades.
- ❖ It thus affects all forms of life for generation to come.



Air Pollution Control

Acts

1948:-Factories Act and Amendment in 1987.

1981:-Air(Prevention and Control of Pollution)Act.

1982:-Air(Prevention and Control of Pollution)Rules.

1982:-Atomic Energy Act.

1987:-Air(Prevention and Control of
Pollution)Amendment Act.

1988:-Motor Vehicle Act.



The Air (Prevention and Control of Pollution) Act, 1981

The Air(Prevention and Control of Pollution)Act, 1981 is a central Act of Parliament, which provides for the prevention and control of air pollution and maintaining the good quality of air.

It also provided for the establishment of the Central and State Boards for the Prevention and Control of Air Pollution.

It is applicable to whole of India except Jammu and Kashmir.



Objectiv

e To provide for the prevention, control and abatement of air pollution.

To provide for the establishment of central and state board with a view to implement the Act.

To confer on the boards the power to implement the provisions of the Act and assign to the boards functions relating to pollution.



Definition Under the Air

~~Act~~ "Air Pollutant" means any solid, liquid or gaseous substance(including noise) present in the atmosphere in such concentration as may be or tend to be injurious to human beings or other living creatures or plants or property or environment.(Sec.2(a))

"Air Pollutant" means the presence in the atmosphere of any air.(Sec.(b))

"Board" means the Central and State Board.(Sec.2(f))

"Member" means a member of the Central Board or a State Board, as the case may be, and includes the

7 Chairman thereof.(Sec.2(l))

Central Pollution Control Board

Central Pollution Control Board constituted under sec.3 of the Water Prevention and Control Act, 1974 was authorized to exercise the powers and performs the functions for the prevention and control of air pollution. (CPCB) of India is a statutory organisation under the Ministry of Environment, Forest and Climate Change. It serves as a field formation and provide technical and financial services.



The functions of Central Board under the Air Act

The main functions of the Central Board, as specified in Section 16 of the Act, shall be:-

Advise the Central Government on any matter concerning the improvement of the quality of air and the prevention, control of abatement of air pollution.

Plan and cause to be executed a nation-wide programme for the prevention and control of air pollution.

Coordinate the activities of the State Boards and resolve dispute among them.

Provide technical assistance and guidance to the State Boards.

Collect, compile and publish technical and statistical data relating to air pollution and measures devised for its

▶ ⁹effective prevention and control.

3 December 2018

The functions of State Boards under the Air Act

The functions of the State Board, as specified in Sec.17, shall be:-

To advise the State Government on any matter concerning the prevention, control or abatement of air pollution.

To collaborate with the Central Board in organising the training of persons engaged or to be engaged in programmes relating to prevention and control of air pollution and to provide mass education programmes relating thereof.

To advise the State Government with respect to the suitability of any premises or location for carrying on any industry which is likely to cause air pollution.

To provide various standards for the emissions of different industrial plants with regard to quantify and composition of emission of air pollutant into the atmosphere.

Penaltie

§ The persons managing industry are to be penalized if they produce emissions of air pollutants in excess of the standards laid down by the State Board.

Whoever is founded to be work against the clauses of the Act or any order or direction issued is punishable with Rs.10,000 in case of three months and if in case offence is continue then extra Rs.5000 is charged for everyday.



The Water (Prevention and Control of Pollution) Act,

1974

The Federal Water Pollution Control Act of 1948 was the first major U.S. law to address water pollution. Growing public awareness and concern for controlling water pollution led to sweeping amendments in 1972. As amended in 1972, the law became commonly known as the Clean Water Act (CWA).

- The need for legislating the act was felt in the year 1962, it was the same year when independent India fought its first war with china. A committee was set up in 1962 to draw a draft enactment for the prevention of water pollution.



Introduction

- The Water Act was enacted by Parliament Act, 1974 purpose to provide for the prevention of control of water pollution and the maintaining or restoring of wholesomeness of water.

As on day, it is applicable in all the states of India.



OBJECTIVES

The water(prevention and pollution control) act 1974, provides for following:

1. Prevention and control of water pollution
 2. Maintaining or restoring the wholesomeness of water; and
 3. Establishment of board for prevention and control of water pollution
- An act to provide for the levy and collection of a cess on water consumed by persons carrying on certain industries and by local authorities, with a view to augment the resources of the central board and the state boards for the prevention and control of water pollution constituted under the water (prevention and control of pollution) act, 1974.



Functions of Central Board (sec. 16)

- The Central Board may establish or recognize a laboratory or laboratories to enable the Central Board to perform its functions under this section efficiently.
- The Central Board may-
 - (a) delegate any of its functions under this Act generally or specially to any of the committees appointed by it;
 - (b) do such other things and perform such other acts as it may think necessary for the proper discharge of its functions and generally for the purpose of carrying into effect the purposes Of this Act.



Functions of the State Board

- Functions of a State Board has been prescribed in section 17
- But the state pollution board have confined themselves only to grant consents to the industries.
- After the legislation of Environment (Protection) Act, 1986 wherein the Standards for Effluent has been prescribed in the EP Rules, the Boards merely mentioned those rules in the conditions of the consent.
- More so Boards are issuing consent with the condition “Zero Discharge” without ensuring whether this condition could be met or complied by the Industry.

